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What to Do About SSRI Withdrawal Mitochondrial Dysfunction

The tenth and final installment in our series on understanding the truth about SSRIs.



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AUG 15, 2025



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So far in this [series](#) we have seen the following:

- That Prozac is a performance-enhancing drug but inferior to exercise as an antidepressant
- That serotonin is a whole-body signaling compound acting primarily outside the brain on cells that are not neurons to control the hypoxia response, which is fundamental to the regulation of energy metabolism
- That SSRIs are primarily mitochondrial drugs rather than “psychiatric” drugs
- That protracted SSRI withdrawal is a form of mitochondrial dysfunction
- That SSRIs can act as mitochondrial poisons, despite their paradoxical ability to also stimulate positive mitochondrial functions
- And that hypobaric hypoxia training is 25 times more effective at promoting psychological resilience than pharmacological antidepressants.

The latter finding is strongly suggestive that depression is a variation of “hypoxia intolerance.”

Many people may not find this easy to grasp because “hypoxia” sounds like you must be getting strangled or drowning under water.

But the hypoxia response is *always* active to some degree in various places in the body as a central means of regulating energy metabolism in a completely healthy human. This is because you always have variation in how much oxygen is reaching various places in your body and what their demands for mitochondrial energy production are. Any relative mismatch between the supply and demand for oxygen will elicit *some* stimulation of the hypoxia response to restore equilibrium.

While some of the effects of SSRI withdrawal are a return of the depression or anxiety that the SSRI was used to treat in the first place, and while some are a result of readjustments in neurotransmitters taking place, the protracted withdrawal that lasts months or years rather than a few weeks is clearly a form of mitochondrial dysfunction that is lasting so long due to the rising of vicious cycles that are locking in the dysfunctional state.

In this tenth and final article in the SSRI series, we brainstorm some things that could be tried to remedy this mitochondrial dysfunction.

This is educational in nature and not medical or dietetic advice. See [terms](#) for additional and more complete disclaimers.

What the Limited Literature Says

The first and most robust defense against SSRI-associated mitochondrial dysfunction is to avoid going on SSRIs in the first place.

The second and next most powerful defense is to stay on SSRIs for as short a time as possible, and to limit the course to less than a year and, if possible, to a few months or even a few weeks. The likelihood of severe and protracted withdrawal increases dramatically with the time spent on the SSRI, and the difference between one year and eight years is likely enormous.

The third defense is to initiate a withdrawal that is as long and drawn-out as possible under the supervision of a very experienced physician.

There is an exception to this: if you've only been on an SSRI for less than eight weeks, you may be best off getting rid of it as fast as possible. This is because somewhere between one and eight weeks is the minimum length of time necessary to make withdrawal a problem. You don't want a slow taper to turn you from someone who didn't need to worry about tapering into someone who has now been on SSRIs for months and needs to worry about each dose-lowering event.

There is some support for using bupropion (Wellbutrin) and PDE5 inhibitors (e.g. Cialis or Viagra) for SSRI-related sexual dysfunction, which may generalize to post-SSRI sexual dysfunction (PSSD). Titrating up [Terra Soul Gelatinized Maca](#) from ¼ teaspoon to 3 tablespoons per day could boost dopamine and norepinephrine and act as a more natural form of bupropion. Titrating up [citrulline](#) from 2 grams to 10 grams per day could boost nitric oxide function and act as a more natural substitute for PDE5 inhibitors. There is anecdotal support for androgen therapy, and my [5 Ways to Boost Testosterone Naturally](#) could come in handy for more natural alternatives.

A [systematic review](#) published earlier this year found that going back on the original SSRI helped in 47% of cases of protracted SSRI withdrawal, while going on other drugs, such as benzodiazepines, pregabalin or propranolol only helped in 18% of cases. Three to six months of cognitive behavioral therapy helped in three out of three cases reported. The GABA-stimulating effects of benzodiazepines and pregabalin might be replicated in natural and more gentle ways with 100-800 milligrams per day of [oral GABA](#), or [chamomile](#).

New Ideas Based on Optimizing Mitochondrial Function

This series has suggested throughout that mainstream progress on this topic is hamstrung by the commitment to the “psychiatric” theory of SSRIs and that real progress can only be made by looking at these drugs through a “mitochondrial” lens.

Here are some ideas to try based on peering at the topic through this lens.

Near Infrared Light

[One case report](#) showed that low-powered laser in the near infrared range on the scrotum and spine of a PSSD patient with no feeling in his genitals led to a 40% increase in penile sensitivity.

[Infrared light](#) acts in part by feeding photons to complex IV (cytochrome oxidase), which improves its ability to facilitate ATP production. Further, near infrared also [structures mitochondrial water](#) to improve its viscosity in a way that facilitates greater production of ATP by ATP synthase.

You can apply therapeutic near infrared light from five to thirty minutes per day (700-1000 nanometers) on each part of the body. For example, [Saunaspace](#) (expensive, whole body), [RedRush Pulse](#) (less expensive, sit by it a lot, turn to face different sides), or [LUMEBOX](#) Red Light (even less expensive but small and you have to apply it to specific areas).

The best source of near infrared light is the sun, but you have to be very careful to manage not getting burned, and these devices allow a greater dose of near infrared without exposure to burning rays.

Recent research suggests that infrared applied to one area in your body can benefit another. For example, when the light is [applied to the chest](#), it improves vision. However, it makes the most sense focus on applying the therapeutic light to the most relevant part of your body, as long as you don't shine it directly in your eyes. For example, try applying it to your head to see if it helps with brain zaps.

Creatine

Creatine is your mitochondria's power grid.

Creatine travels through the cell orders of magnitude faster than ATP, so it is the primary way you spread the energy produced by the mitochondria throughout the rest of the cell.

As I covered in [Your Cells Are Starving For Creatine](#), *everyone* needs to optimize their creatine status. If you aren't consuming one to two pounds of animal flesh per day, or if you aren't consuming 500 grams per day each of quinoa and tofu, you almost certainly need to supplement with creatine.

The success story I shared [earlier in the series](#), who reported at 8.5 months that "I've still been pretty amazed how effective your protocol has been. On the whole I feel like a whole different person from a year ago. It's wild" was using creatine as a central part of his protocol.

He had to start at 100 *milligrams* of creatine for a long time due to the twitches it caused him. He managed these with my strategies in [Handling Creatine Side Effects](#).

You can start with a five-day loading dose of 20 grams per day followed by a 3-gram-per day maintenance dose, but if you have adverse effects you should start at a dose that doesn't cause them, even if that means reverting to something as low as 100 milligrams. The loading dose helps you achieve muscle saturation quickly. You can also achieve muscle saturation by using 3 grams a day for a month. Creatine should be spread out into three or four doses per day taken before meals (not after meals), after exercise, or on an empty stomach to facilitate maximal absorption. After achieving muscle saturation, you can titrate up at one gram per day per week as high as 20 grams per day, but should stop at the point of maximal benefit, which might be around 3-5 grams, could be 7-8 grams, and for some people might be 20 grams.

The [Mitome](#) test had showed that this client had impaired transport of methyl groups into the mitochondria, and since creatine supports methylation, this may have made him respond especially well to the creatine. However, everyone needs to optimize their creatine status so this is a safe and fruitful path to embark on without testing first.

Melatonin

The animal evidence covered in [Melatonin Is Your Mitochondria's Guardian Angel](#) suggests that secretion of melatonin at night by the pineal gland is a major source of mitochondrial melatonin throughout the body as cells clear it from the blood by taking it up and transporting much of it into their mitochondria. Less predictable stimulation of the hypoxia response during periodic hypoxic stress supplies the remainder of the melatonin by driving circulating *serotonin* into mitochondria where it is *converted* to melatonin.

SSRIs likely deplete mitochondrial melatonin by depleting serotonin at a whole-body level and in most specific tissues outside the brain and gut, and by blocking the entry of serotonin into the mitochondria for melatonin synthesis.

The depletion of mitochondrial melatonin could lead to a vicious cycle where mitochondrial function is damaged from constant hypoxic signaling that respiration is unnecessary and a constant state of melatonin deficiency, thus hurting the ability of mitochondria to synthesize melatonin from serotonin, leaving them unable to lift themselves up by their bootstraps once the SSRI is taken away. Supplemental melatonin may therefore help break the vicious cycle.

[Human trials of melatonin supplementation](#) have used 5-10 milligrams per day for up to three months, 10-50 milligrams per day for up to one month, and 100-400 milligrams per day for up to two weeks, and 1000 milligrams on one single day. Case reports have used as many as 1600 milligrams per day for 24 days in individual patients. High-dose melatonin increases the risk of drowsiness, headache, and dizziness by 40%, but it does not increase the risk of severe adverse effects.

Overall, high-dose melatonin appears safe for a limited period of time, with the safety of using 10-50 milligrams for up to a month demonstrated rather robustly.

Most studies use melatonin at night, though some have used it in the day. I think it's clear that melatonin should be used before bed so as not to disrupt the cyclical rise of plasma melatonin at night and its fall during the day. If you supplement before bed,

the plasma level will rise more than usual, but this will also be rapidly cleared by cells that take it up and can deliver it to their mitochondria.

The use of melatonin in this manner should be thought of as a short-term repletion of the mitochondrial melatonin pool. This can be done with [10-50 milligrams of melatonin](#) taken before bed for up to one month. You should be able to stop this cold turkey with no problems, but if you notice any decline in sleep quality during withdrawal you could try tapering it slowly.

Strategies to Improve Mitochondrial Turnover

SSRIs can dysregulate mitochondrial turnover by promoting mitochondrial biogenesis (the creation of new mitochondria) through specific serotonin receptors and by promoting mitophagy (breakdown of damaged mitochondria) through direct activation of the sigma-1 receptor.

SSRIs with low sigma-1 activation are likely to overstimulate mitochondrial biogenesis relative to mitophagy, while those with high sigma-1 activation are likely to overstimulate mitophagy relative to mitochondrial biogenesis. *Withdrawal* from these SSRIs could do the opposite.

So, withdrawal from fluvoxamine and sertraline, which are strong sigma-1 activators, would be expected to cause a loss of sufficient mitophagy, while withdrawal from paroxetine, citalopram, and escitalopram would be expected to cause a loss of mitochondrial biogenesis. Fluoxetine would be in the middle.

However, this also will depend on an individual's genetic variation in all of the relevant receptors.

Mitophagy and mitochondrial biogenesis are regulated by the supply and demand for mitochondrial energy metabolism. In general you need *both* demand for new mitochondria and supply of the substances needed to produce mitochondria.

Exercise is an excellent way to stimulate the demand side of mitochondrial biogenesis, where induction of an energetic deficit by the exercise can increase mitophagy and subsequent resting and refeeding can increase mitochondrial biogenesis.

Cognitive behavioral therapy could be seen as a form of exercise since it will create demand for mitochondrial energy production in all of the neural networks that support the cognitive and behavioral changes being stimulated in the same way bodily exercise does this for neural networks that promote strength and for energy production within muscle itself.

Your exercise program should be designed relative to your capacity. If you are chronically fatigued and feel incapable of serious exercise, take what you *can* do, cut it down to 60-70%, and begin an exercise cycle where you aim to slowly do more over time so that by the end of twelve weeks you can do more than you had been able to at the start. If you see no change to your capacity, you may need to work harder. If you see yourself getting weaker and more fatigued as you do more work, you need to reduce your volume, frequency, duration, or intensity, or select an easier exercise, and rest more. Find the right pace of rest and recovery and take off bites you can chew.

My [fasting-feeding reset](#) is designed around a cyclical promotion of mitophagy in the fasting state and mitochondrial biogenesis in the fed state. It uses exercise and supplements to amplify each state on a cyclical basis. This is a well balanced approach. However, given the possibility that SSRI withdrawal has left you with a pronounced *imbalance*, it is important to pay close attention to your body's responses and avoid excessive stress. If you find your resilience declining rather than improving, deemphasize the portion of the cycle that seems to be causing the most bodily stress. For example, you may want to greatly reduce the stimulation of the fasting part of the cycle while indulging more in the refeed or vice versa.

While various supplements can be leveraged to globally increase mitophagy or mitochondrial biogenesis, this risks making a major imbalance worse.

I recommend testing [Mitome](#) before doing this, which will tell you whether you are more imbalanced in one direction or the other and will put the relevant supplements directly into your protocol.

Another approach is **intermittent hypobaric hypoxia training**. As I covered in [This Is 25x More Effective Than Antidepressants—And It's Not a Drug](#), animal studies indicate this is a powerful promoter of psychological resilience and it may be addressing the root cause of depression far more directly than SSRIs do.

While I strongly promote everyone engaging in some degree of hypoxic stress as part of their health regimen, this is another case where [Mitome](#) testing would provide value. Too much hypoxia can hurt the respiratory chain, and idiosyncratic differences in the hypoxia response can underly the different responses to different SSRIs. Mitome can see whether your hypoxia regimen is excessive or just right and the protocol will help you adjust it.

5-HTP for Certain SSRIs

SSRIs with poor sigma-1 activation like **paroxetine, citalopram, and escitalopram** deplete serotonin even in the brain, and recovering from these might warrant supplementation with **200 to 900 milligrams per day of [5-HTP](#)**, with or without [Seeking Health B6 as P5P](#) 25 milligrams once a week to once a day.

Fluoxetine is in a gray area, whereas sertraline and fluvoxamine not likely to do this.

Serotonin synthesis is likely to be restored with adequate protein (at least 1.2 grams per day per kilogram of ideal bodyweight), adequate micronutrients, and the proper assortment of stressors covered above like exercise, cognitive behavioral therapy, and intermittent hypoxia.

However, the ability to support 5-HTP synthesis is complicated by a large number of nutritional cofactors, and while it does not occur in the mitochondria, it does depend on mitochondrial energy metabolism.

The ability to convert 5-HTP into serotonin is far more simple and only requires vitamin B6.

Serotonin depletion is less likely to lock in a vicious cycle than mitochondrial melatonin depletion, which is why this is considered a possible adjunct for certain SSRIs rather than a general principle applied to withdrawal from any of them.

Runner Up: Natural Sigma-1 Activation

A few more things deserve some mention but their roles aren't quite clear.

Chronic activation of the sigma-1 receptor by fluvoxamine or sertraline could leave it chronically underactivated during withdrawal. This receptor is never supposed to be chronically activated, so what SSRIs are doing to it is completely unnatural.

We know that it should be activated periodically during stress, so this is another case where healthy stressors like exercise and intermittent hypoxia become important. We don't know what naturally activates it, but DHEA-sulfate, choline, myristic acid, and sphingolipids have all been suggested. Choline can be obtained from liver and egg yolks; myristic acid from coconut oil and butter; and sphingolipids from the fatty portions of a wide variety of foods, with unique compositions found in organ meats, fish, shellfish, mushrooms, whole grains, and fermented foods.

Many Relevant Nutrients

As with any other health problem, you need to make sure you are hitting all of your nutrient targets. [Track your diet in Cronometer](#) to make sure you are hitting them on paper, and run the [Comprehensive Nutritional Screening](#) using the [Cheat Sheet](#) for interpretation to make sure your regimen is working to support your body's unique needs.

Everyone Is Different

Everyone has variation in the specific strengths and weaknesses of their mitochondria, and assaults on the mitochondria that precipitate the onset of chronic mitochondrial dysfunction will reveal these deficits. Testing the mitochondria in the dysfunctional state can provide insight into what the limiting bottlenecks are, and can therefore add a highly individualized layer to your protocol that makes it specific to correcting your own dysfunction. [Mitome](#) tests your mitochondrial function with a cheek swab and delivers a personalized protocol that you could integrate with the SSRI-specific strategies outlined above.

Share Your Experience in the Comments!

If you have SSRI withdrawal mitochondrial dysfunction and have tried some of these strategies, let us know what worked and didn't work in the comments.

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Discussion about this post

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Gillian Tobin 1h ...

Would this also apply to SNRI’s? How does norepinephrine affect mitochondrial function? I am about to start a slow hyperbolic taper off of Cymbalta. I’ve been on it for 11 years. I’ve been on various psych meds 20 years. I’m trying to get as much information as possible.

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Nathaniel Nathanson 2h *Edited* ...

Thanks for your working on this Chris, it means a lot to those of us suffering. You talk a lot about serotonin in this article - some of us are suffering from finasteride, Accutane, and lions mane - does serotonin still apply for those people?

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